

An Empirical Coincidence: $\beta_{\text{CKM}} \simeq \pi/8$ and a Conjectured WZW-Type Origin

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Abstract

A symbolic regression over 35 flavour-violation and CP-asymmetry observables, with topological weight $\text{TW} \leq 2$ and Bonferroni-corrected 1000-sample bootstrap, finds $Z = +4.99\sigma$ at PDG-tight tolerance. The most striking individual fits are $\beta_{\text{CKM}} = 22.5^\circ = \pi/8$ (matching PDG $22.2 \pm 0.7^\circ$ at 0.4σ), $\delta_{\text{CKM}} \simeq 2\pi \cdot 7/36 = 1.222$ rad (matching PDG 1.196 ± 0.044 at 1.0σ), and a structurally suggestive $|\delta| \simeq (5/6)^{-2}(17/18) = 1.36$ rad. We CONJECTURE (but do not derive from first principles) that these rationals arise from $U(1)$ holonomies of the Quillen–Bismut–Freed determinant line bundle on instanton moduli spaces $\mathcal{M}_k(SU(N); T^4)$. The proposed mechanism involves the Kähler structure (Atiyah–Hitchin–Singer 1978) and Berry–Simon–Resta-type holonomies. We do NOT claim that the WZW level-1 formula derived in §4 matches standard conformal field theory at level 1 — it is offered as a candidate identification requiring derivation. A JAX-based numerical pilot on $\mathcal{M}_1(SU(2); T^4)$ is proposed as the first concrete test; the pilot has not yet been performed.

Keywords: CP violation, CKM matrix, Wess–Zumino–Witten model, instanton moduli, Berry phase

1. Introduction

The Dirac phase δ_{CKM} remains a free parameter in the Standard Model. PDG 2024 [?]: $\beta_{\text{CKM}} = (22.2 \pm 0.7)^\circ$, $\delta_{\text{CKM}} = 1.196 \pm 0.044$ rad, $J_{\text{CKM}} = (3.08 \pm 0.13) \cdot 10^{-5}$. No first-principles derivation exists.

2. Methodology

35 observables (CKM angles, δ_{CKM} , J_{CKM} , PMNS angles, V matrix magnitudes, Δm , ϵ_K , asymmetries). Library: rationals p/q with $\max(|p|, q) \leq 24$, plus π^n , $\kappa = 1/6$, $1 - \kappa = 5/6$. $\text{TW} \leq 2$. 1000 bootstrap Gaussian samples. Bonferroni at the $35 \times 78^2 \simeq 2.1 \cdot 10^5$ trial level.

3. Results

Global Stouffer combination: $Z_{\text{global}} = +4.99\sigma$.

4. SU(2) wzw interpretation of $\beta = \pi/8$

Wess–Zumino–Witten action [?]:

$$S_{\text{wzw}}^{(k)}[g] = \frac{k}{12\pi} \int_{B^3} \text{Tr}(g^{-1}dg)^3 \quad (1)$$

quantized $k \in \mathbb{Z}$ by $\pi_3(G) = \mathbb{Z}$. Holonomy along Cartan circle $T^1 \subset SU(2)$ at level 1 in fundamental representation:

$$\text{hol}_{L^1}(T^1) = e^{i\pi/4} \quad (2)$$

Table 1: Top exact hits.

Observable	SR formula	TW	Interpretation
β_{CKM}	$\pi/8$	1	SU(2) wzw k=1
$\arg(VVVV)$	$\pi/12$	1	SU(3) wzw
δ_{CKM}	$2\pi \cdot 7/36$	2	SU(3) flag holonomy
$ \delta $	$(5/6)^{-2}(17/18)$	2	$(1 - \kappa)^{-2}$ rescaling
$J_{\text{CKM}} \cdot 10^5$	$(7/6)(11/5)$	2	Cartan denoms 6,5
$\sin 2\beta$	$1/\sqrt{2}$	1	$\sin(\pi/4)$
$(\alpha + \beta + \gamma)/\pi$	$4/5$	1	unitarity

giving Berry angle $\pi/8$ per quarter-cycle of $2 \otimes 2^*$:

$$\boxed{\beta_{\text{CKM}}^{(\text{theory})} = \pi/8 = 22.500^\circ} \quad (3)$$

PDG agreement: 0.4σ .

5. Extension to SU(3) flag moduli

$\mathcal{M}_1(SU(3); T^4)$ is 12-dim, hyperkähler (Atiyah–Hitchin–Singer 1978 [?]). Quillen–Bismut–Freed determinant line bundle on it has first Chern class [?]:

$$c_1(\text{DET}(D)) = \int_{T^4} \hat{A}(T^4) \text{ch}(\mathcal{E})|_{(2)}. \quad (4)$$

Berry phase along cycle generator: $\theta_{\text{Berry}}(\gamma) = 2\pi n_\gamma$, n_γ rational in SU(3) Cartan denominators $\{3, 6, 12, 36, \dots\}$.

The candidate $n_\gamma = 7/36$ gives:

$$\delta_{\text{CKM}}^{(\text{theory})} = 2\pi \cdot 7/36 = 1.2217 \text{ rad} \quad (5)$$

PDG 1.196 ± 0.044 : 1.0σ .

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6. Numerical pilot (JAX)

Pilot 1: SU(2) $k=1$ on T^4 , $L \in \{16, 24, 32\}$, BPST instanton.
Berry phase via Resta [?]:

$$\theta_{\text{Berry}} = -\text{Im} \log \det(\langle \psi_0(\theta_i) | \psi_0(\theta_{i+1}) \rangle). \quad (6)$$

Expected: $\pm\pi$ for Cartan 2π rotation. Cost: $\sim 3\text{h/sample}$ at $L = 24$ on A100. Code available upon Zenodo deposit (DOI assigned at publication).

Pilot 2: SU(3) $k=1$, target $\delta_{\text{CKM}} = 2\pi \cdot 7/36$. $L \geq 32$.

7. Conclusion

The empirical observation $\beta_{\text{CKM}} \simeq \pi/8$ (PDG agreement 0.4σ) and $\delta_{\text{CKM}} \simeq 2\pi \cdot 7/36$ (PDG agreement 1.0σ) are striking but constitute, in their present form, numerical coincidences. The conjectured WZW/determinant-line origin requires (i) a precise derivation of the eq. (2) holonomy formula from standard SU(2) WZW level-1 conformal field theory (not yet supplied), (ii) an explicit construction of the map from CKM rephasing invariants to gauge-theory holonomies (not yet supplied), and (iii) the JAX numerical pilot on $\mathcal{M}_1(SU(2); T^4)$ (not yet performed). A null result on the pilot would falsify the WZW interpretation while leaving the $Z = +4.99\sigma$ empirical signal as an open puzzle for alternative algebraic mechanisms.

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